



Interview Outcome Classification with BERT

Fine-tuning ruBERT-base to automatically
classify recruitment call outcomes into
five categories — a data analyst case study.

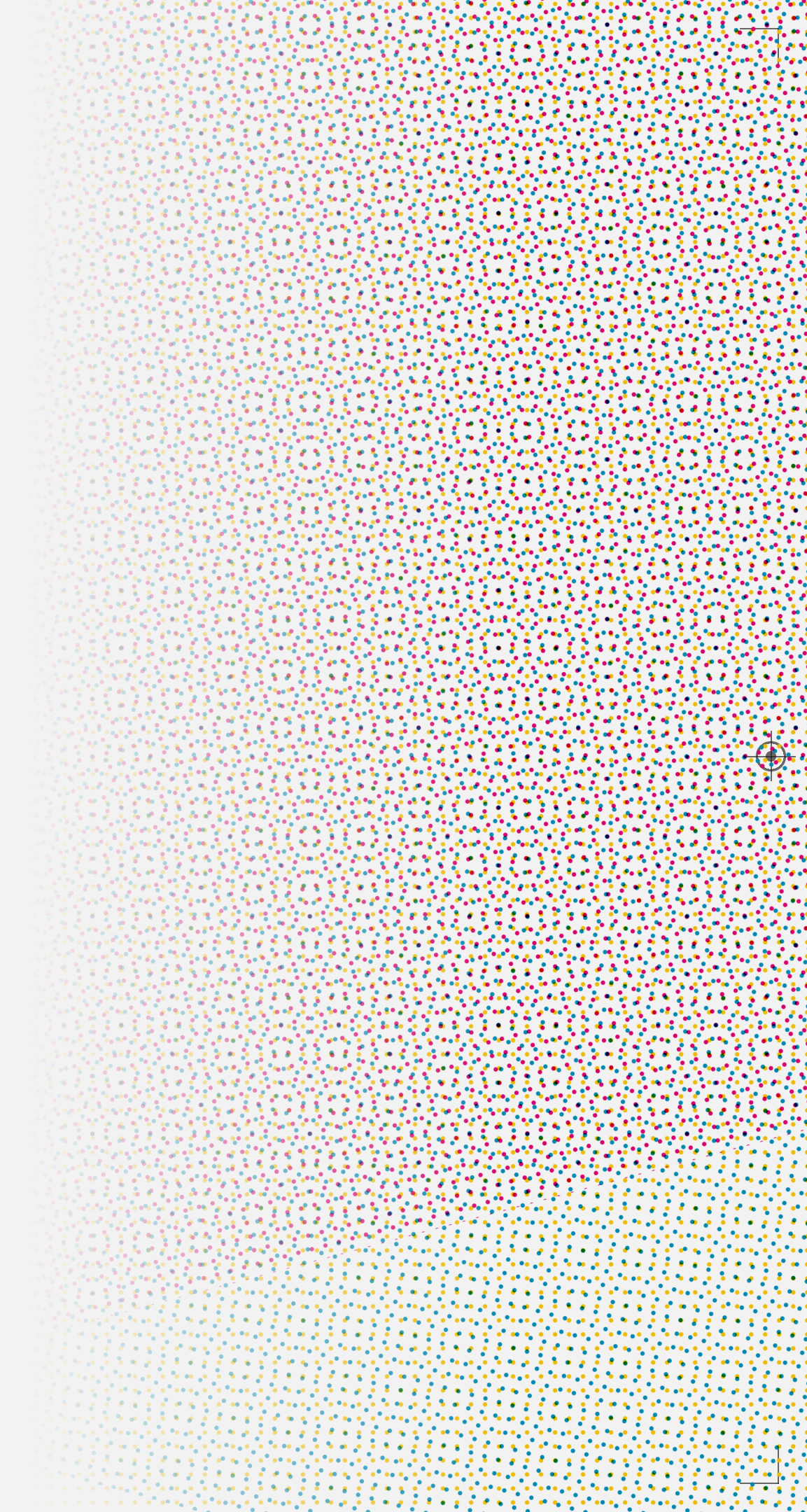


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The problem

01



Recruiter call outcomes are labelled by hand

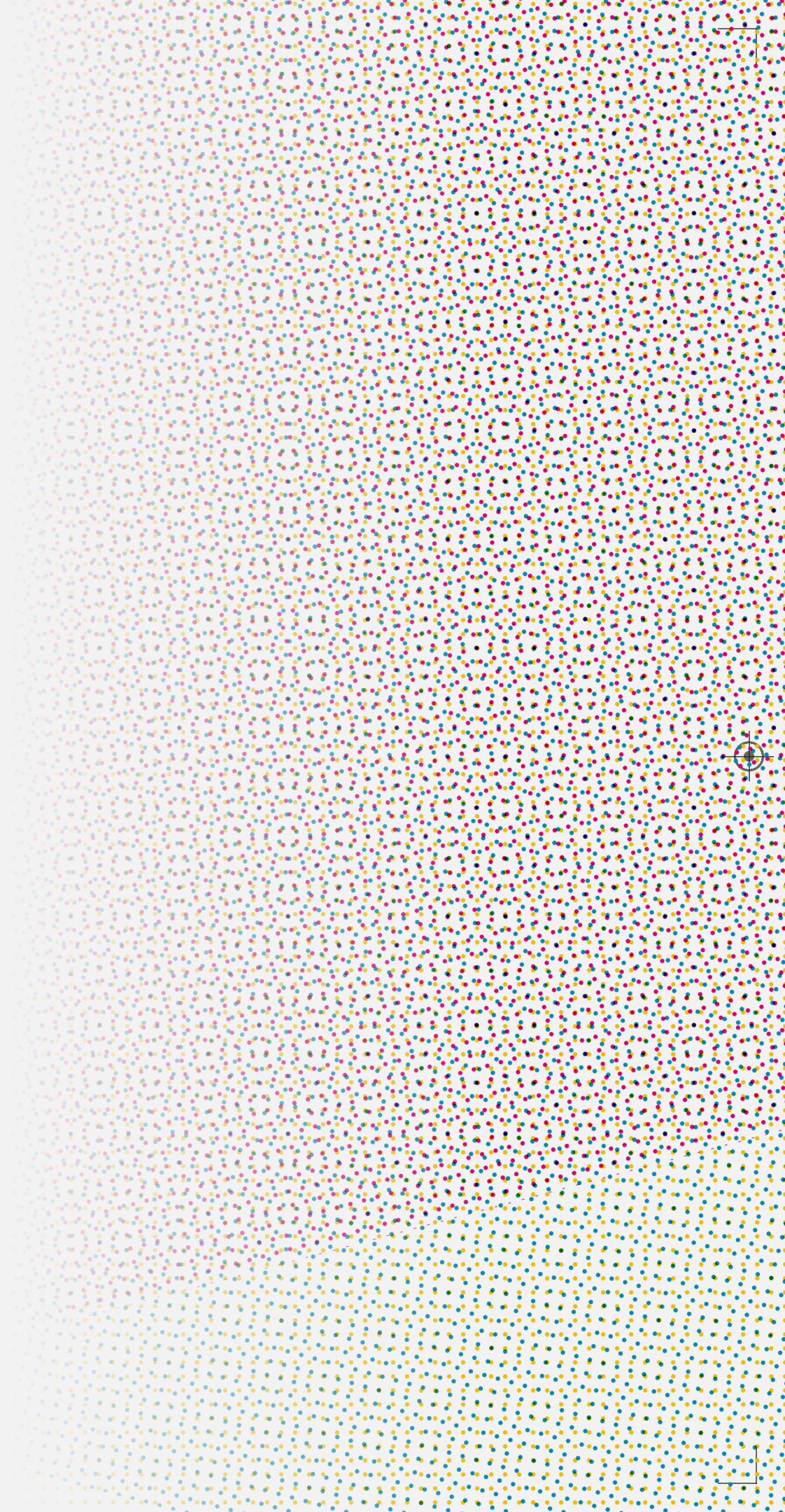
- Recruitment teams conduct hundreds of phone interviews a week, and every call outcome is logged manually.
- Manual labelling is slow and inconsistent — the same outcome gets tagged differently across recruiters.
- Goal: train a Russian-language model to classify a call transcript into a structured outcome, enabling downstream analytics, reporting and recruiter performance tracking.

Five outcome labels

LABEL	MEANING
0	Candidate declined
1	Recruiter declined the candidate
2	Candidate agreed
3	Next steps were discussed
4	Candidate actually started the job

Calls that failed for technical reasons carry no signal about interview outcome and are excluded from training.

Modeling approach

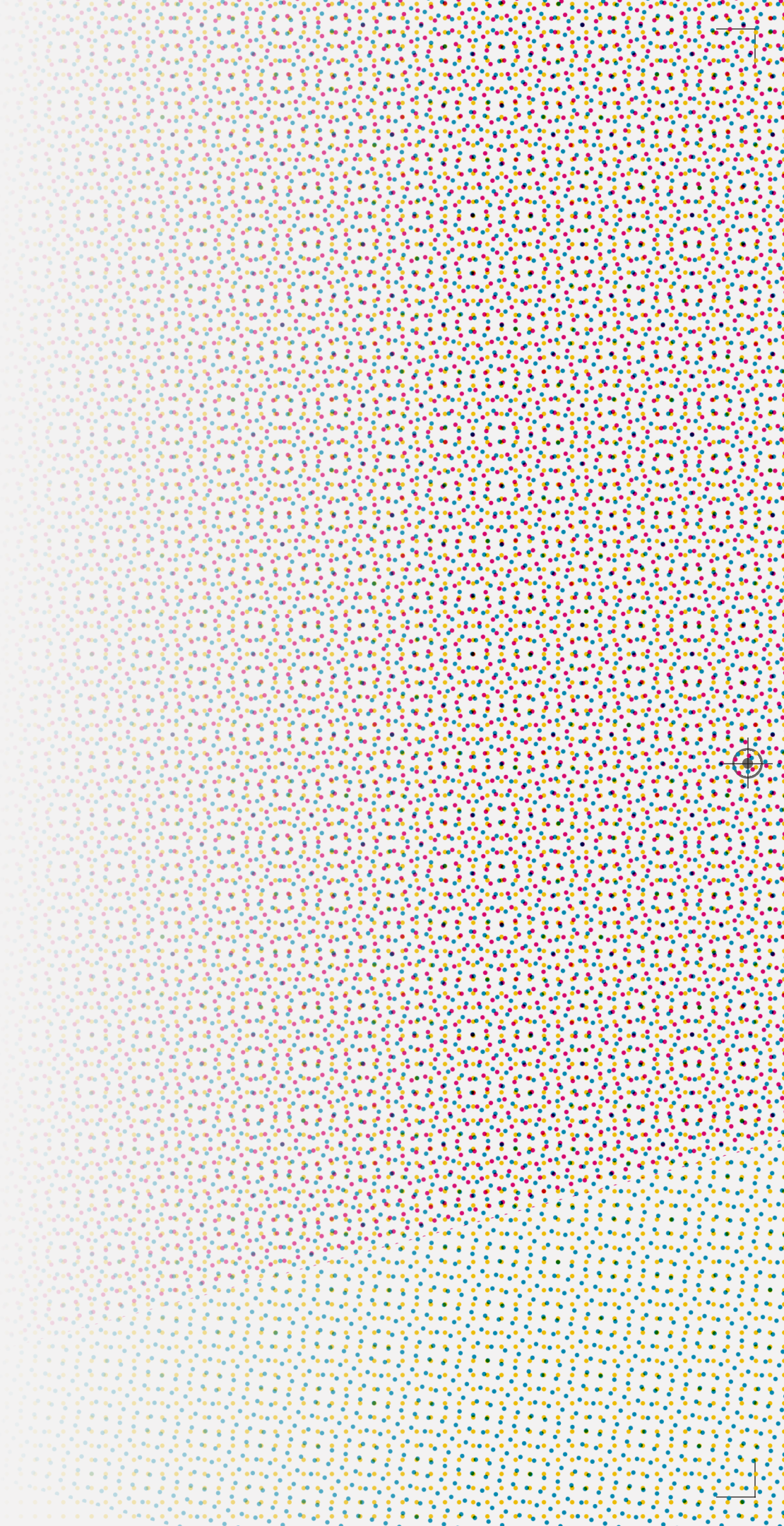


Fine-tuning ruBERT-base

PARAMETER	VALUE
Base model	ai-forever/ruBert-base
Fine-tuning head	BertForSequenceClassification — dropout + linear on [CLS]
Tokenisation	WordPiece, max length 512, dynamic padding
Optimiser	AdamW, lr = 1e-6
Batch size	4
Epochs	3
Train / val split	80 / 20, stratified by label

Results

03





870%

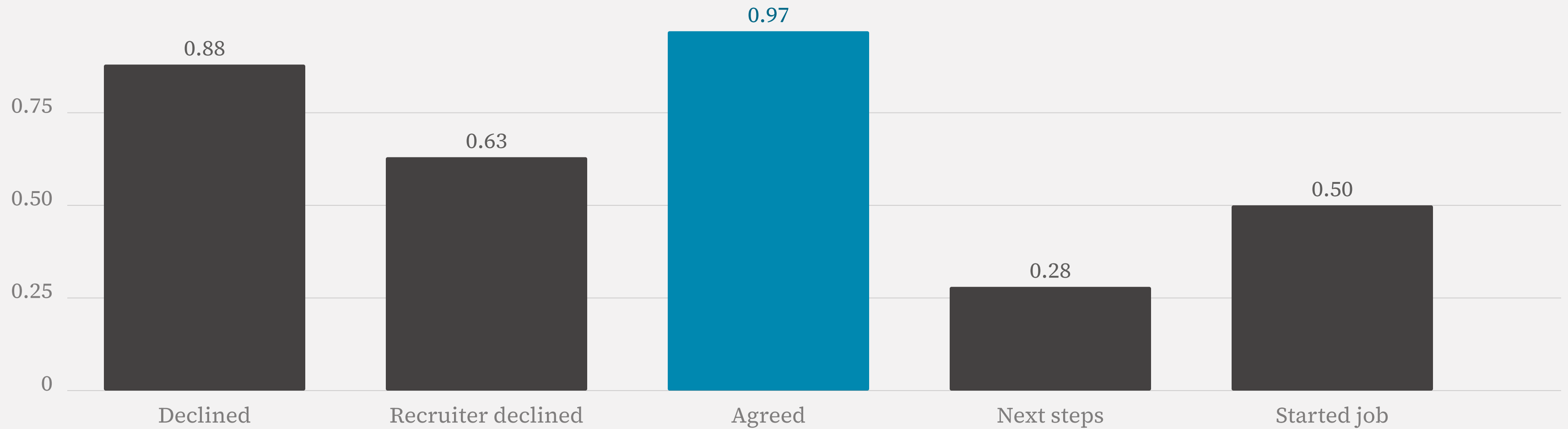
Overall accuracy on a held-out validation set of 2,129 labelled calls — weighted F1 of 0.86 across the five outcome classes.

Performance by class

CLASS	PRECISION	RECALL	F1	SUPPORT
Candidate declined	0.86	0.90	0.88	467
Recruiter declined candidate	0.86	0.49	0.63	77
Candidate agreed	0.96	0.98	0.97	1,313
Next steps discussed	0.35	0.23	0.28	124
Candidate started job	0.47	0.54	0.50	148

Macro average F1 0.65, weighted average F1 0.86 — the imbalance between classes drives the gap between the two.

F1 score by class



Candidate agreed dominates the training data and is classified most reliably; Next steps discussed is the hardest class — it is frequently confused with Candidate started job.

Confusion matrix

ACTUAL \ PREDICTED	0	1	2	3	4
0 — Declined	420	6	41	0	0
1 — Recruiter declined	29	38	10	0	0
2 — Agreed	24	0	1,288	1	0
3 — Next steps	5	0	2	28	89
4 — Started job	12	0	6	50	80

Labels 3 and 4 confuse each other in both directions — both involve future-oriented language, and the boundary between them is semantically thin.

Limitations & future work

Class imbalance

Recruiter declined (77 samples) and Next steps discussed (124) are dwarfed by Candidate agreed (1,313). Class-weighted loss or oversampling would likely lift both classes' recall.

Label ambiguity

Next steps discussed and Candidate started job sit close in meaning. Clearer annotation guidelines, or merging the two classes, would remove much of that confusion.



Thank you

ruBERT-base fine-tuned for five-way
interview outcome classification —
full notebook and README available
on request.